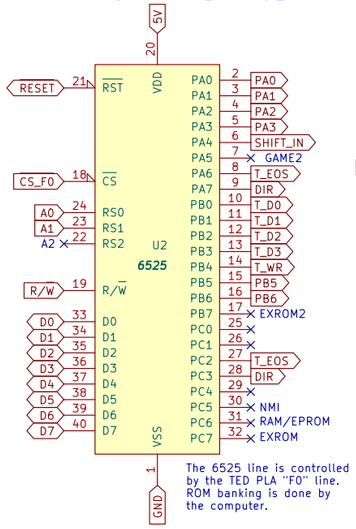
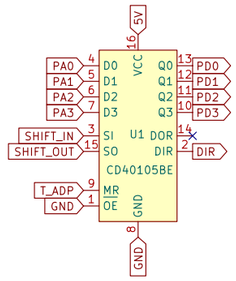
## TRI-PORT INTERFACE

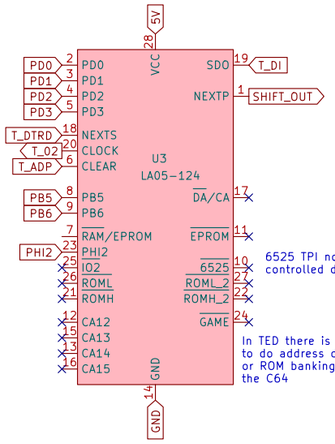



The 6525 line is controlled by the TED PLA "F0" line. ROM banking is done by the computer.

This is the 6525 Tri-Port Interface chip. It was used in a few Commodore products like the CBM-II computers, the C64 IEEE cartridge, Amiga CDTV, and A570. They are rare, but it might be possible to find one or perhaps the design might be able to use an INTEL 8255.




## GI LOGIC ARRAY



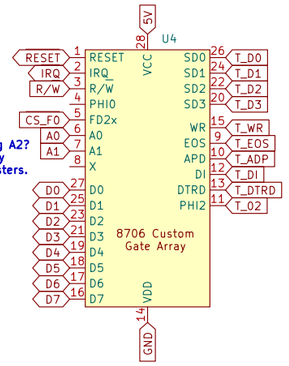

6525 TPI now controlled directly

In TED there is no need to do address decoding or ROM banking like on the C64

This is the custom gate array only found in the Magic Voice. You will not find a replacement but there is a recreation project that hopefully works well enough to use here.

**This circuit is similar to what is included in the V364's custom Gate Array.**

## CUSTOM GATE ARRAY

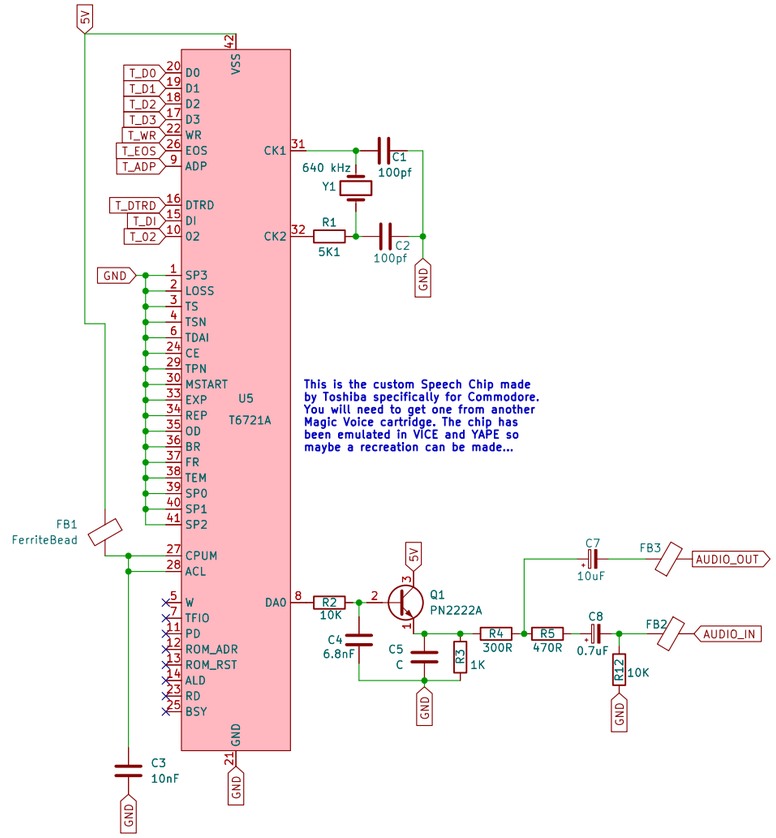



Missing A2? So only 4 registers.

This is the 8706 Custom Gate Array. It was developed specifically for the V364 and when the machine was cancelled Commodore re-purposed and adapted the design for use in the C64 using the MOS TPI and off-the-shelf GI Logic Array.

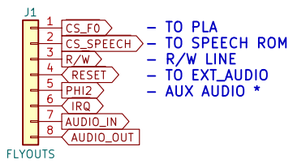
NOTES: Need to investigate the difference in software of the TPI vs the 8706 and mapping of the registers.

## SPEECH CHIP




This is the custom Speech Chip made by Toshiba specifically for Commodore. You will need to get one from another Magic Voice cartridge. The chip has been emulated in VICE and YAPE so maybe a recreation can be made...

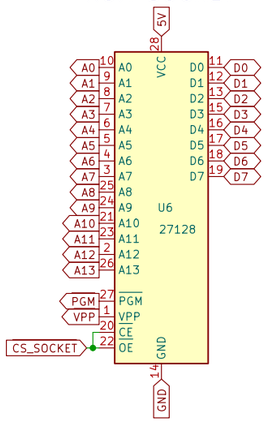
## FLYOUTS




- TO PLA
- TO SPEECH ROM
- R/W LINE
- TO EXT_AUDIO
- AUX AUDIO *

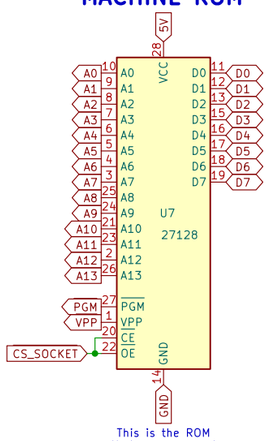
* Could be from an internal SID or FM board.

## ROM SOCKET



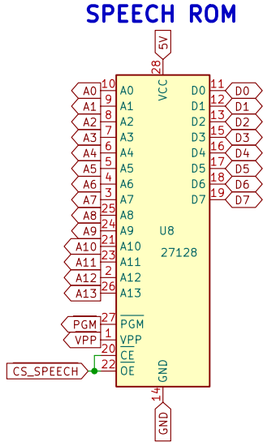

This is the socket that this Dev board connects to.

## MACHINE ROM




This is the ROM that was removed from the socket.

## SPEECH ROM




This is the SPEECH ROM from a real V364.

## DESCRIPTION

This is a Speech Dev Board for 264 Series.
It uses either the C64 Magic Voice (MV) chips, or the V364 chips.
With MV it uses the TPI, Gate Array, and Speech chip from an original MV unit. It also has a socket for the 8706 Custom Gate Array recreation.

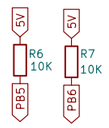
The board connects to one of the TED ROM sockets and that ROM is relocated to this board. The V364 Speech ROM is installed on this board (or a patched ROM if required).

On the TED motherboard the PLA "F0" line and the Speech ROM chip select must be connected to this board.

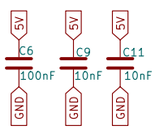
Audio output should be wired to the EXT_AUDIO line of the Cartridge Port.

**CREATED: 2026-10-06**
**UPDATED: 2026-10-08**

## PULL-UPS




## BYPASS CAPS




# Speech Dev Board

Commodore Business Machines
**by Steve J. Gray**
Sheet: /
File: DevMV.kicad_sch

***Title: Internal Magic Voice***

| Size: A3 | Date: 2026-10-06 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |

**WORK IN PROGRESS!!!**